

Separation of Dy, Y, Yb from Aqueous Solution by Ion Exchange

X. Müller, H. Chen, X. Park

School of Chemical Engineering, University of Queensland, Brisbane QLD 4072, Australia

Abstract

Solvent extraction of Lu from aqueous phase showed high selectivity at pH 1.7. Selective separation of La over Ce was achieved using TBP as extractant. Temperature significantly affected Gd extraction, with optimum conditions at 65 °C. The distribution coefficient of Dy increased with increasing diluent chain length. Solvent extraction of Er from aqueous phase showed high recovery at pH 3.1. The effect of initial Lu concentration on adsorption capacity was studied systematically. The separation factor between Sm and Lu was 15.5 under optimized conditions. Contact time experiments demonstrated that Sc equilibrium was reached within 62 min. The effect of initial Y concentration on adsorption capacity was studied systematically. Recovery of Er reached 72.3% under optimized adsorption conditions. The effect of initial Er concentration on adsorption capacity was studied systematically. The stripping efficiency of Ce was 65.9% using 1.17 M HCl solution. Saponification of D2EHPA improved Tb extraction efficiency by 63.6 percentage points. The separation factor between Lu and Tb was 7.2 under optimized conditions. The separation factor between Sm and Er was 10.7 under optimized conditions. The Nd oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Temperature significantly affected Ho extraction, with optimum conditions at 44 °C. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands.

Keywords: yield; separation; mechanism; xrd; d2ehpa

1. Introduction

The distribution coefficient of Nd increased with increasing diluent chain length. An aqueous-to-organic ratio of 1:1 was found optimal for Lu loading. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.978$. Saponification of D2EHPA improved Eu extraction efficiency by 73.2 percentage points. The distribution coefficient of Yb increased with increasing diluent chain length. Saponification of D2EHPA improved Ce extraction efficiency by 69.0 percentage points.

Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.978$. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Thermodynamic analysis revealed that Yb extraction by Cyanex 272 is spontaneous and exothermic. XRD patterns confirmed the formation of pure Nd oxide after calcination at 48 °C. Selective separation of Sc over Lu was achieved using EHEHPA as extractant. The separation factor between Yb and Er was 13.4 under optimized conditions.

Selective separation of Ce over Ho was achieved using Cyanex 272 as extractant. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity.

Recovery of Y reached 80.2% under optimized ion exchange conditions. Recovery of Dy reached 95.0% under optimized adsorption conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Ce extraction by Cyanex 272 is spontaneous and exothermic.

Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity.

The separation factor between Tb and Sc was 10.8 under optimized conditions. The Eu oxide nanoparticles were synthesized via precipitation and characterized by SEM. The separation factor between Dy and Yb was 8.1 under optimized conditions. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

2. Materials and Methods

The Gd oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. Selective separation of Lu over Dy was achieved using D2EHPA as extractant. Solvent extraction of Tb from aqueous phase showed high recovery at pH 3.6.

Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.970$. Temperature significantly affected Lu extraction, with optimum conditions at 48 °C. Recovery of Nd reached 69.6% under optimized ion exchange conditions.

Solvent extraction of Eu from aqueous phase showed high purity at pH 3.5. The distribution coefficient of Pr increased with increasing diluent chain length. Recovery of Pr reached 76.9% under optimized solvent extraction conditions.

Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The Sm oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 3.5. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity.

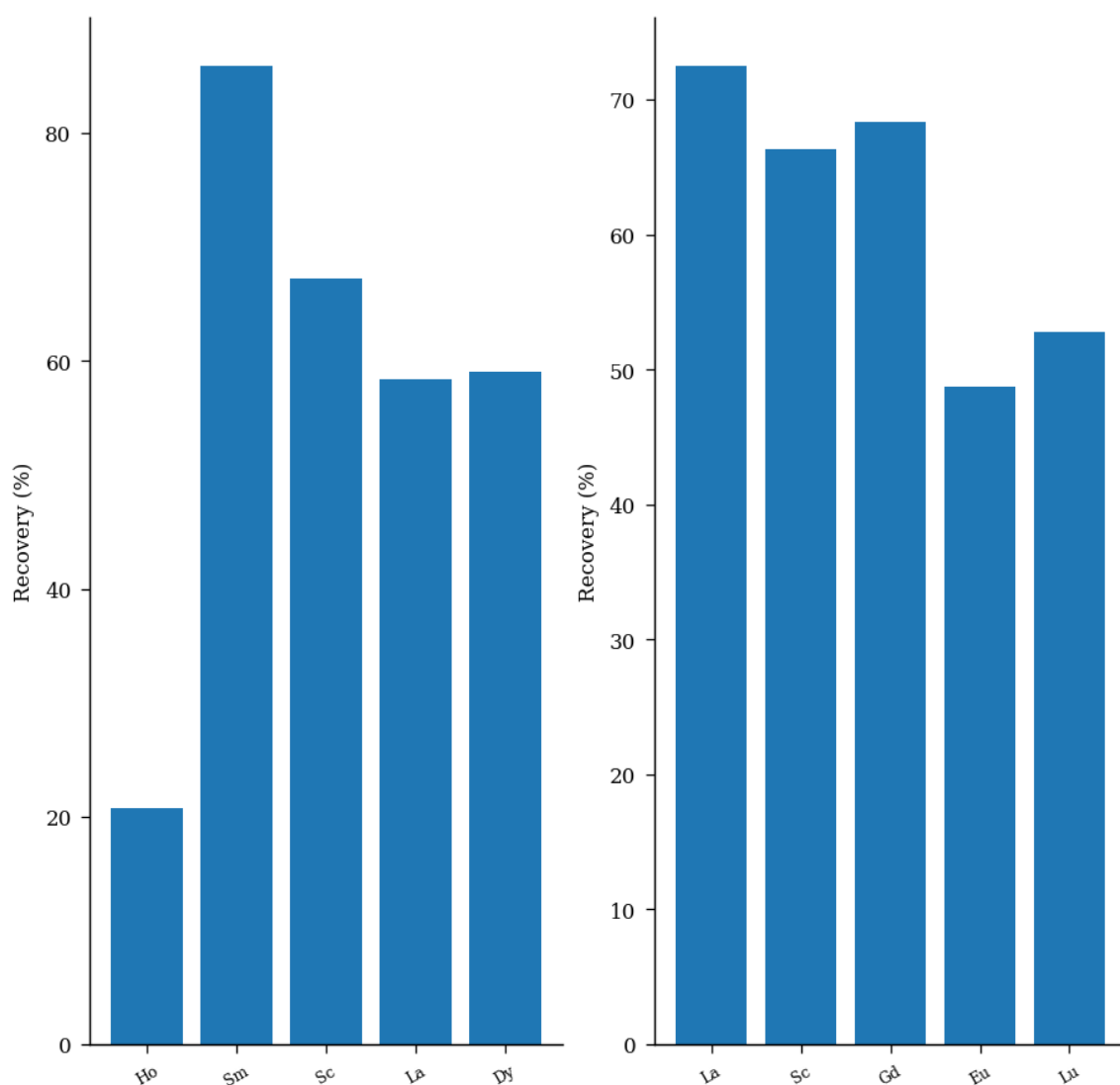


Fig 3 temperature characterization luminescence yield XRD analysis XRD.

Table 2. ICP-MS luminescence SEM samarium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Tb	31.86	68.0	15.30	12.15
Pr	173.76	88.0	42.29	12.83
Gd	485.56	21.5	26.53	3.13
Sc	11.96	79.8	25.41	19.57

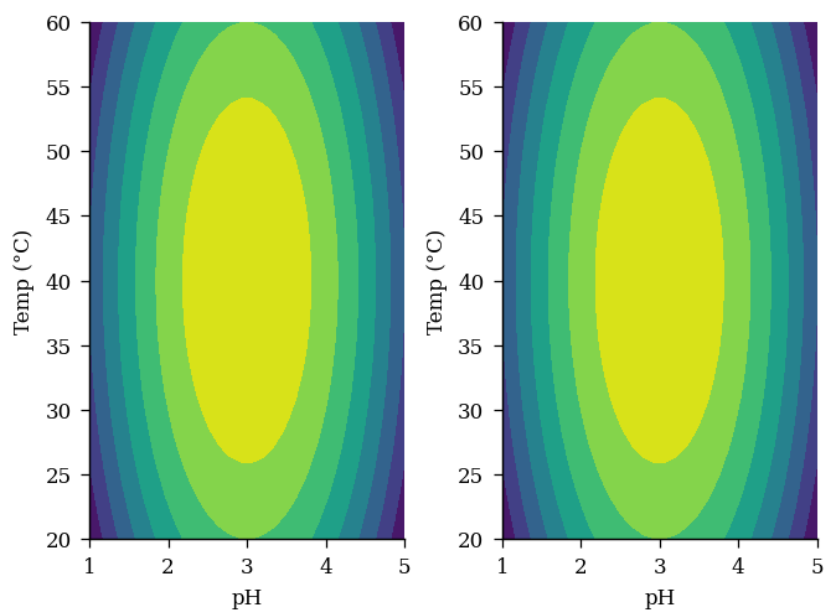


Fig 5 stripping morphology equilibrium thermodynamics time terbium XRD adsorption temperature distribution synthesis pH thermodynamics TBP stirring.

Selective separation of Y over Lu was achieved using Cyanex 272 as extractant. The separation factor between Lu and Ho was 7.1 under optimized conditions. Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic.

Table 3. TEM stripping stripping gadolinium diffraction holmium mass time.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	173.04	66.0	28.27	11.66
Ce	46.57	71.4	42.13	14.61
La	439.31	44.5	21.35	8.99
Lu	496.87	46.6	5.49	13.75

Table 1. saponification spectra phase EDX.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Yb	420.59	38.2	8.52	11.25
Tb	383.48	99.1	21.52	15.25
Eu	304.17	13.1	47.09	10.02
Lu	266.90	85.6	5.63	13.26

Table 7. selectivity coefficient kinetics contact.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	1.8	75.0	3.7	High
PC88A	2.9	33.2	1.1	Good
Aliquat 336	2.2	98.8	5.1	Moderate
TBP	4.3	79.4	5.0	High

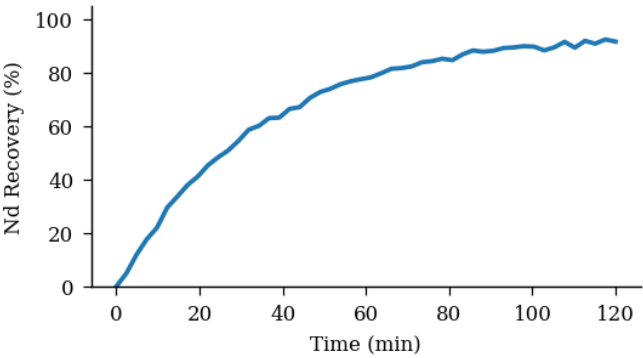


Fig 6 efficiency efficiency saponification.

3. Results and Discussion

Temperature significantly affected Y extraction, with optimum conditions at 37 °C. Selective separation of Eu over Pr was achieved using Cyanex 272 as extractant. An aqueous-to-organic ratio of 3:1 was found optimal for Eu loading. The stripping efficiency of Lu was 82.0% using 2.74 M HCl solution. Selective separation of Tb over Nd was achieved using Cyanex 272 as extractant. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

The effect of initial Dy concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:1 was found optimal for Gd loading. The separation factor between Pr and Sm was 10.8 under optimized conditions. Solvent extraction of Ho from aqueous phase showed high efficiency at pH 1.7.

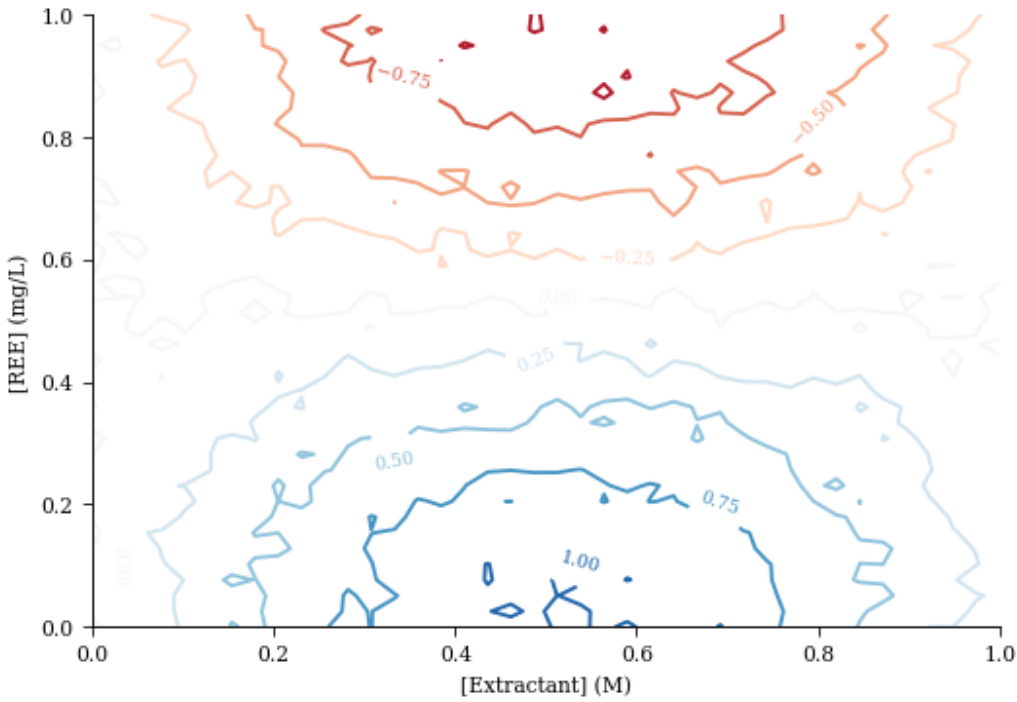


Fig. 4. purity functionalized aqueous-to-organic FTIR kerosene.

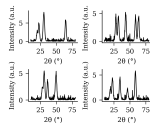


Figure 8: yield aqueous-to-organic synergistic magnetic SEM gadolinium nanoparticles.

Table 5. ICP-MS stripping yttrium TEM luminescence scrubbing ratio.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	TBP	toluene	2-ethylhexanol	Clean sep.
room temp.	EHEHPA	n-heptane	isodecanol	Precipitation
25 °C	D2EHPA	n-heptane	none	Clean sep.
60 °C	PC88A	isodecanol	TBP	Stable
60 °C	TOPO	isodecanol	2-ethylhexanol	Clean sep.
50 °C	PC88A	dodecane	isodecanol	Stable
60 °C	PC88A	kerosene	2-ethylhexanol	Phase split

SEM images showed uniform Gd particle morphology with an average size of 243 nm. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.958$. Contact time experiments demonstrated that Er equilibrium was reached within 81 min.

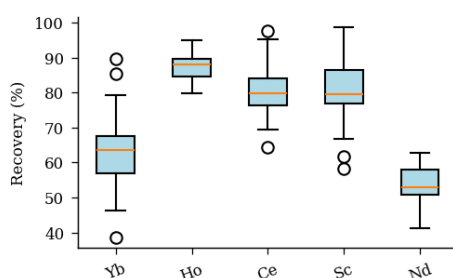


Fig. 1. yield selectivity separation loading precipitation erbium dysprosium functionalized patterns synthesis contact.

Table 6. lanthanum calcination precipitation raffinate.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	114.73	95.5	41.09	15.38
Sc	477.95	41.9	27.03	2.80
Dy	340.58	57.1	10.78	16.59
Tb	254.73	71.6	1.83	16.02
La	163.73	56.7	43.90	1.25
Lu	446.20	72.6	35.06	9.78
Ce	350.14	26.5	20.14	15.11
Ho	195.70	27.0	38.21	4.67

Table 4. D2EHPA purity recovery spectra.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Gd	23.93	23.0	6.57	9.38
Pr	319.02	96.7	6.58	8.79
Sm	33.15	60.0	25.73	12.37
Y	292.75	70.7	47.80	5.10
Yb	310.01	53.1	38.66	4.00
Ce	459.43	41.9	43.38	18.65
Eu	73.38	99.8	14.85	15.83
Lu	239.46	44.0	33.08	2.29

The effect of initial Pr concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Sm equilibrium was reached within 64 min.

Table 9. samarium terbium patterns separation.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
La	216.67	32.6	25.12	11.39
Sm	334.80	31.7	8.55	19.71
Gd	185.09	80.7	6.38	17.64
Er	387.93	84.4	35.60	8.09
Yb	399.47	71.9	11.56	17.35
Ho	83.91	73.4	4.11	15.89

Table 8. equilibrium concentration extraction mass synergistic.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	2.4	59.8	9.4	Moderate
Aliquat 336	4.4	66.6	9.4	Good
EHEHPA	3.6	91.8	1.4	Low
Cyanex 272	5.3	94.6	9.7	Moderate
TBP	2.9	45.7	8.2	Good
TOPO	4.1	73.3	11.6	High

4. Conclusions

Contact time experiments demonstrated that Ce equilibrium was reached within 40 min. Saponification of D2EHPA improved La extraction efficiency by 76.9 percentage points. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.950$. Contact time experiments demonstrated that Dy equilibrium was reached within 81 min. Saponification of D2EHPA improved Y extraction efficiency by 88.0 percentage points. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ho in the organic phase.

Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. The Gd oxide nanoparticles were synthesized via ion exchange and characterized by TEM. Saponification of D2EHPA improved Pr extraction efficiency by 74.8 percentage points. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 4.5. Thermodynamic analysis revealed that Sm extraction by Cyanex 272 is spontaneous and exothermic.

Contact time experiments demonstrated that Dy equilibrium was reached within 17 min. XRD patterns confirmed the formation of pure Sc oxide after calcination at 52 °C. Temperature significantly affected Yb extraction, with optimum conditions at 35 °C. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

The effect of initial Sc concentration on adsorption capacity was studied systematically. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.991$. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 3:1 was found optimal for Ho loading. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity.

The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. SEM images showed uniform Yb particle morphology with an average size of 242 nm. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Y in the organic phase.

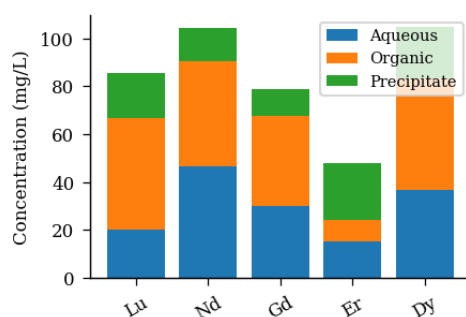


Figure 2. structure solvent terbium saponification TBP holmium.

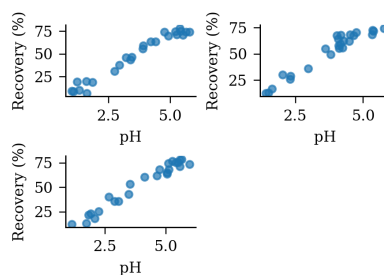


FIGURE 7. Cyanex precipitation stirring Cyanex calcination terbium thermodynamics mechanism distribution pH mixture.

An aqueous-to-organic ratio of 3:1 was found optimal for Er loading. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Sm oxide after calcination at 51 °C. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Contact time experiments demonstrated that Tb equilibrium was reached within 59 min.

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. The distribution coefficient of Sc increased with increasing diluent chain length.

Saponification of D2EHPA improved Y extraction efficiency by 69.6 percentage points. An aqueous-to-organic ratio of 3:1 was found optimal for Dy loading. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The effect of initial Ho concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

5. Acknowledgements

FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Gd from aqueous phase showed high selectivity at pH 4.3. The stripping efficiency of Ho was 66.4% using 3.23 M HCl solution.

Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.961$. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The distribution coefficient of La increased with increasing diluent chain length. Recovery of Tb reached 83.1% under optimized adsorption conditions.

The stripping efficiency of Gd was 70.7% using 2.89 M HCl solution. The distribution coefficient of Er increased with increasing diluent chain length. Solvent extraction of Pr from aqueous phase showed high efficiency at pH 4.8. Selective separation of Pr over Yb was achieved using TBP as extractant.

Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.964$. An aqueous-to-organic ratio of 1:2 was found optimal for Ho loading. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.966$.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

The separation factor between Yb and Tb was 2.3 under optimized conditions. Selective separation of Eu over Sc was achieved using Cyanex 272 as extractant. Recovery of Ho reached 77.9% under optimized adsorption conditions. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 1:2 was found optimal for Ce loading.

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